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Paper Code: TCS 851

End Semester Examination 2025

Name of the Course: B.Tech.

Semester: 8TH

Paper Name: Storage Networks

Time: 3 Hours

Maximum Marks: 100

Note:

- (i) Answer **all the questions** by choosing **any Two of the sub questions**.
- (ii) Each question carries 10 marks.

(2X10=20 Marks)

CO1/CO2/CO3

- Q1. a. A large company is considering a storage infrastructure- one that is scalable and provides high availability. More importantly, the company also needs performance for its mission-critical applications. Which storage topology would you recommend (SAN, NAS, IPSAN) and why?
- b. i) What is HBA? Discuss the benefits of using multiple HBAs on a host.
ii) Why do formatted disks have less capacity than unformatted disks?
- c. Explain briefly the layers of FC protocol stack.

(2X10=20 Marks)

CO2/CO3/CO4

- Q2. a. Why isn't Fibre Channel used often for the internal drives inside system cabinets? List the four interconnects from slowest to fastest performance for storage networking applications.
- b. What is symmetric storage virtualization? Write its advantages and disadvantages.
- c. Explain storage virtualization at file level, with a necessary diagram.

(2X10=20 Marks)

CO3/CO4/CO5

- Q3. a. What would you consider while choosing serial or parallel data transfer in a DAS implementation? Explain your answer and justify your choice.
- b. Explain the process of data recovery in case of a drive failure in RAID 5.
- c. Why is RAID 0 not an option for Data Protection and high Availability?

(2X10=20 Marks)

CO3/CO5/CO6

- Q4. a. Write a short note on:

- 1. Hot Swap
- 2. Hot spare
- 3. RAID Controller

b. What is zoning? Discuss a scenario where hard zoning is preferred over soft zoning.

c. What is FC-SAN? Discuss the advantage of FC-SW over FC-AL.

(2X10=20 Marks)

CO3/CO5/CO6

- Q5. a. What is Network Attached Storage? Briefly describe the components of NAS. Write a note on Integrated NAS and Gateway NAS.

b. Explain IPSAN technologies. Describe some of the data storage applications that could benefit from IPSAN solution.

c. Explain the factors involved in measuring disk drive performance.

